

Test Cases

White box testing (Internal logic and structure testing)--

Index	Test Steps	Test Data	Expected o/p	Actual o/p	Status	Note
1	Text preprocessing	Valid essay text	Tokens and Word count generated accurately	-	Not executed	NA
2	Grammar model execution	Essay with grammatical errors	Low score	-	Not executed	NLP model Working
3	Scoring logic	Essay i/p	Accurate score calculation	-	Not executed	Scoring formula verified
4	Feedback generation	Short essay	"Essay too short"	-	Not executed	
5	Save essay into DB	User_id, essay	Essay stored with id	-	Not executed	DB exists and verified
6	Save feedback into DB	Essay_id, score, feedback report	Feedback is accurately linked to parent essay	-	Not executed	NA
7	Duplicate handling	Same email	Error handled	-	Not executed	Integrity constraint

Black box testing (i/p o/p testing)—

Index	Test Steps	Test Data	Expected o/p	Actual o/p	Status	Note
1	Essay i/p	Valid essay text	Essay accepted and processed	-	Not executed	NA
2	Essay i/p	Empty string	Error – No essay entered	-	Not executed	NA
3	Essay submission	Essay with grammar errors	Feedback with the errors flagged/highlighted	-	Not executed	NLP model working
4	Scoring system	Valid essay	Score generated	-	Not executed	NA
5	Feedback generation	Essay i/p	Meaningful feedback	-	Not executed	NA
6	Progress comparison	Multiple scores	Performance trend and analysis	-	Not executed	NA
7	Submission history	Multiple essays	Past essays retrievable	-	Not executed	NA

